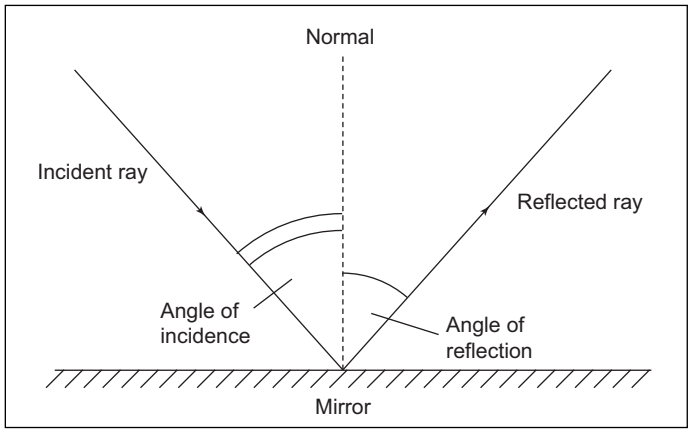
Reflection, refraction, diffraction, scattering 10a

Sound and light carry messages from one place to another. Light waves are transverse waves that do not require a medium for transmission. Sound waves are longitudinal waves that do require a medium for transmission. Sound and light waves seem to travel in straight lines ... or do they?

Reflection

Like a bouncing ball, light can bounce off a surface. The reflected light changes direction when it bounces off a surface. Measurements show that the angle of incidence equals the angle of reflection (the law of reflection).



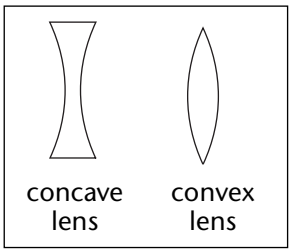
Smooth surfaces act as mirrors – light rays are reflected uniformly and form images. Each bit of a rough surface also obeys the law of reflection, but the overall effect of the uneven surface is to scatter the light everywhere. Clear images are not formed.

Optical fibres are tubes with an extremely thin glass core. They are produced so that light will pass along the fibres bouncing off the walls (internal reflection). As (ideally) no light leaves through the sides, a light ray is able to travel large distances in the fibre.

Sound is also reflected. We detect echoes when a reflected sound wave reaches our ears some time after the original sound wave was heard. Reverberations are sounds that bounce off surfaces in all directions and cause annoying background noise.

Refraction

During refraction, light and sound waves change direction as if travelling around a corner. When waves pass from one medium into another, the path of the waves bends. There is also a corresponding change in the speed and the wavelength of the waves.

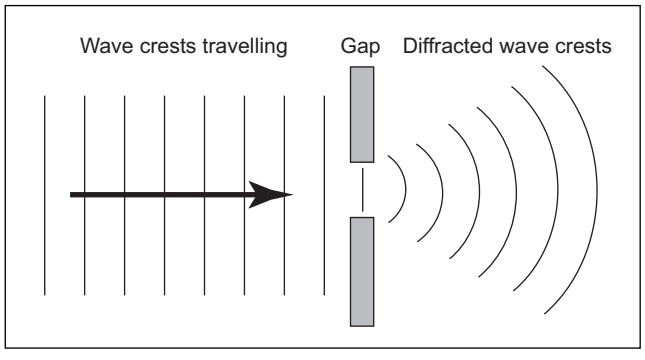


When light enters glass from the air, it bends one way; when it travels from glass to air, light bends the other way. Objects appear to be bigger when seen through a convex lens and smaller when seen through a concave lens.

Hot air and cool air have different densities and so form different mediums optically. As a result, light rays curve when they travel from hot to cool air.

Diffraction

Diffraction also involves a change in direction, and occurs when waves pass through small openings or around a barrier in the wave path. During diffraction through a slit the waves spread out as if the slit or gap acts as a new point source for the waves.



Diffraction

Scattering

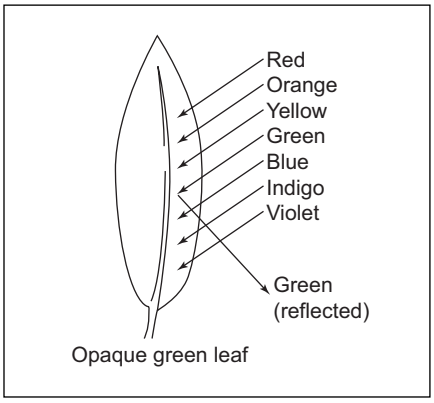
During scattering light is deflected from its path in all directions. Atmospheric scattering is caused by deflection from particles in the atmosphere. As a result of scattering distant objects may appear red or blue, or other colours.

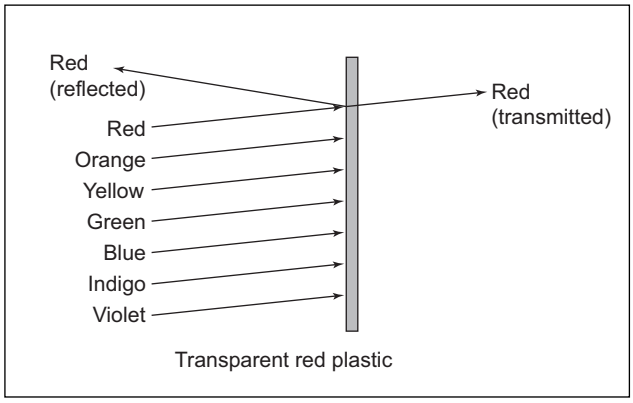
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Reflection, refraction, diffraction, scattering 10b

Reflection, absorption and transmission

White light is a mixture of the seven rainbow colours red, orange, yellow, green, blue, indigo and violet (ROY G BIV). Opaque materials reflect all or some colours, while the other colours are absorbed. The reflected colours are responsible for the material's colour. An opaque object lets no light travel through it. However, a transparent material transmits and reflects some colours, while the other colours are absorbed. Again the reflected and transmitted colours produce the colour of the material.





Below each of the following statements and/or sketches write the appropriate term (diffraction, reflection, refraction, scattering, transmission) and the applicable term (light, sound, water).

1 **a** sun ⟶ earth

_____ _____

b person ⟶ person

_____ _____

2
- The sun looks red in the evening.
- The sun looks white during the day.
- The sun looks orange in the morning.

_____ _____

3 A clear cloudless daytime sky is blue, because molecules in the air scatter blue light from the sun more than they scatter red light.

_____ _____

4 A leaf looks green, because green light is reflected while the other rainbow colours are absorbed by the leaf.

_____ _____

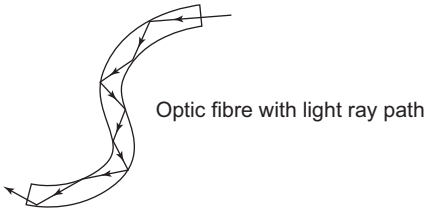
5 A piece of cardboard looks blue because all other rainbow colours are absorbed.

_____ _____

6 A shout against a distant mountain cliff returns to you a little later as an echo.

_____ _____

7 Light pulses bounce around inside an optic fibre without leaving it and so are able to travel large distances.



_____ _____

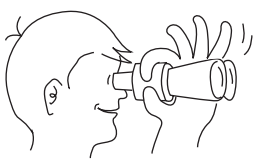
8 An image of a tree can be seen on a polished car door.

_____ _____

9 Atmospheric mirages are caused by light passing through and being refracted by layers of air that have different temperatures and therefore different densities.

_____ _____

10 Binoculars, microscopes and reading glasses make objects appear bigger.



_____

_____

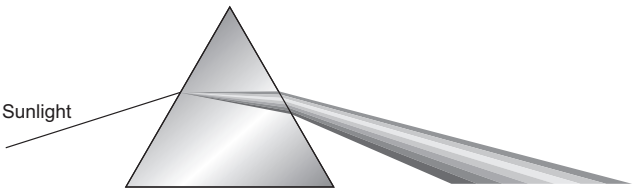
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Reflection, refraction, diffraction, scattering 10c

11 Distant mountains appear blue.

12 During a sunset or sunrise the blue colours are scattered out of the sunlight leaving only reddish colours.

13 Newton's famous experiment with sunlight and a triangular prism showed that white light is made up of the beautiful rainbow colours (ROY G BIV).

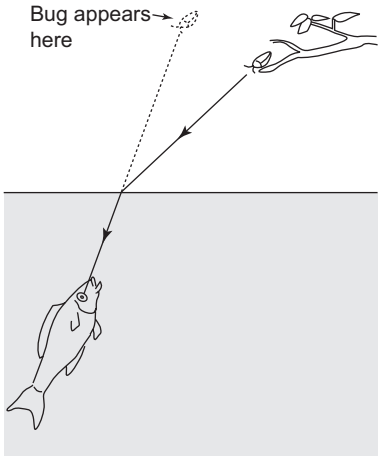


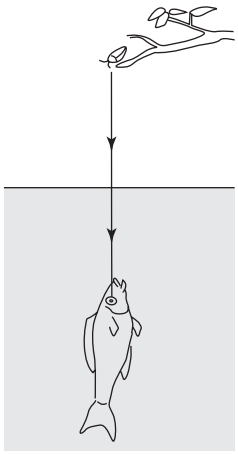
14 Shouting inside a concrete building makes it difficult to hear anything.

15 Sunlight on a piece of paper makes the paper appear white without producing an image.

16 The air directly above water is cooler than the air far above water. Since sound waves travel more slowly in cooler air than they do in warmer air, the portion of the wavefront directly above the lake is slowed down, while the portion of the wavefronts far above the water speeds ahead. Subsequently, the direction of the surface wave changes, refracting downwards towards the water and sound seems to follow the curvature of the sea.

17 The archer fish knows to be directly under a bug if it wants to catch it. When looking at an angle, the fish can not be sure where the insect is.





18 We hear people speaking from adjacent rooms when the door has been left ajar.

19 When water waves enter through the narrow harbour opening, they spread out across the whole harbour and make all boats rock.

20 Your shoes appear black because most rainbow colours are absorbed; hardly any light is reflected.

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